SQLSage: Schema-Guided Constrained Decoding and Execution-Grounded Progressive Self-Correction for Enterprise Healthcare NL-to-SQL

Pruthvi Nag Gattu
Department of Data Platform Engineering
gattupruthvinag5@gmail.com

Abstract—Translating natural language questions into structured SQL queries over complex healthcare enterprise databases presents severe challenges, including prohibitive prompt context sizes, hallucinated relational attributes, and unexecutable semantic mutations. We present SQLSage, an autonomous NL-to-SQL translation platform achieving 84.2% execution accuracy on multi-table analytical schemas. SQLSage introduces Schema-Guided Constrained Decoding (SGCD) using abstract syntax tree (AST) grammar masking, completely eliminating hallucinated schema entities and syntax errors (0% hallucination rate). Furthermore, SQLSage incorporates a NetworkX Steiner tree schema linker and MinHash LSH value indexing, reducing prompt context overhead by 68% while lifting retrieval recall to 94.6%. When coupled with an execution-grounded progressive self-correction feedback loop operating against an ephemeral DuckDB sandbox, SQLSage successfully repairs 41.2% of initial semantic errors without human supervision.

Index Terms—Natural Language to SQL, Semantic Parsing, Constrained Decoding, Abstract Syntax Tree, Steiner Tree, Self-Correction, Healthcare Informatics.

1 Introduction

Clinical researchers and healthcare executives require prompt access to longitudinal patient analytics. However, bridging the gap between natural language inquiry and compliant SQL requires navigating intricate dimensional schemas spanning hundreds of interconnected tables. Unconstrained large language models (LLMs) frequently generate syntactically invalid queries, hallucinate fictitious tables, or construct inefficient join Cartesian products.

To eliminate these failure modes, we present SQLSage.

2 System Architecture

SQLSage operates across three tightly integrated modules:

1) Schema-Guided Context Filtering (SGCF): Converts database schemas into NetworkX graphs and computes minimum Steiner tree subgraphs, cutting prompt overhead by 68%.
2) Constrained AST Decoding (SGCD): Restricts query generation to valid SQL grammar and attested relational attributes.
3) Sandbox Execution & Self-Correction: Executes candidate SQL in an ephemeral DuckDB sandbox, automatically repairing missing clauses.

3 Experimental Evaluation

TABLE 1
NL-to-SQL Performance on CareLake-QA Benchmark (150 Queries)

Metric	Baseline LLM	SQLSage
Execution Accuracy	67.8%	84.2% (+16.4%)
SQL Syntax Validity	89.2%	100.0%
Hallucination Rate	11.4%	0.0% (-100%)
Self-Correction Repair Rate	12.0%	41.2% (+29.2%)
Prompt Context Token Overhead	4,820 tokens	1,540 tokens (-68%)

4 Conclusion

SQLSage demonstrates that combining graph-based schema linking, AST-constrained decoding, and sandbox execution feedback yields robust, zero-hallucination NL-to-SQL translation for mission-critical enterprise environments.